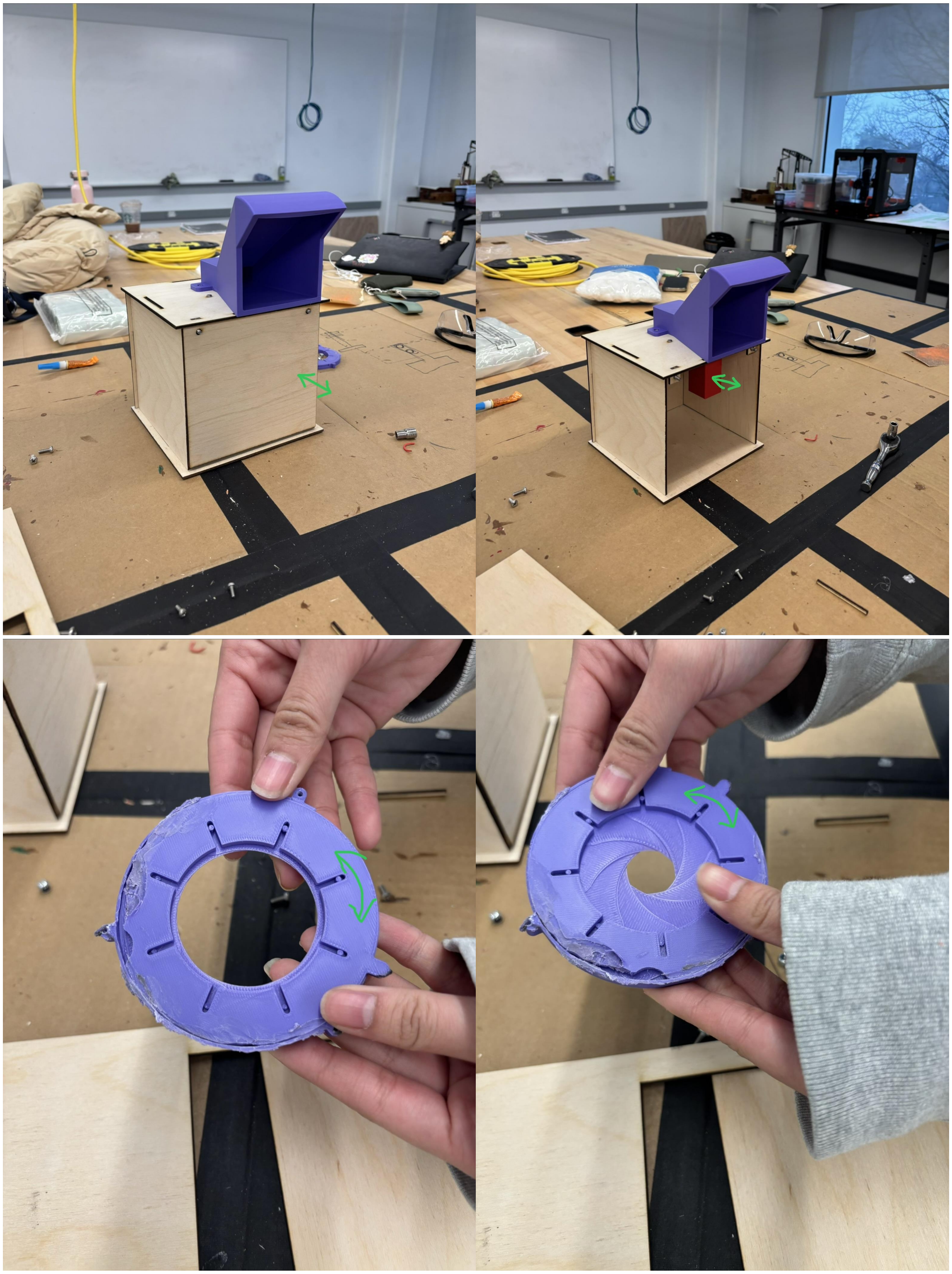
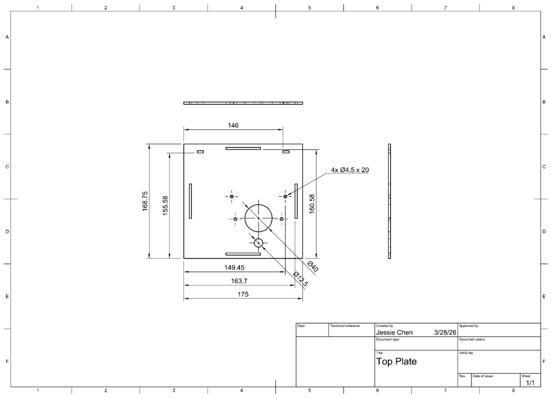
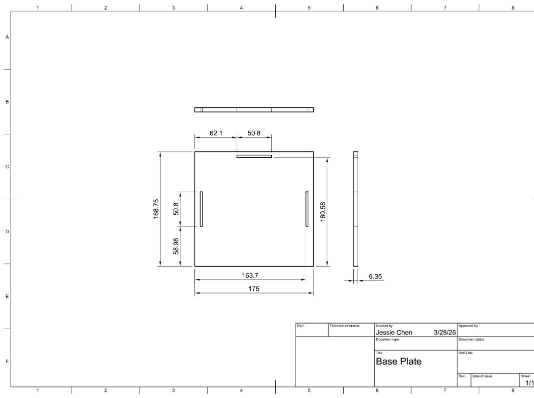
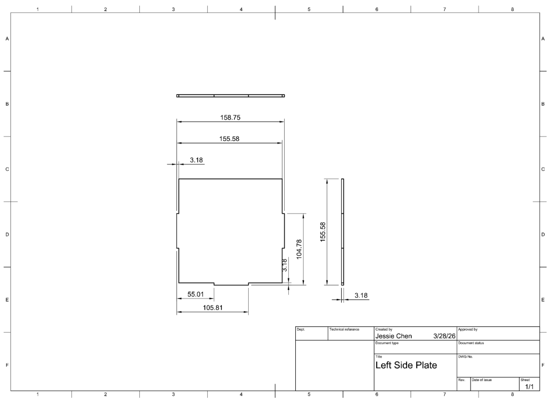
Design Documentation

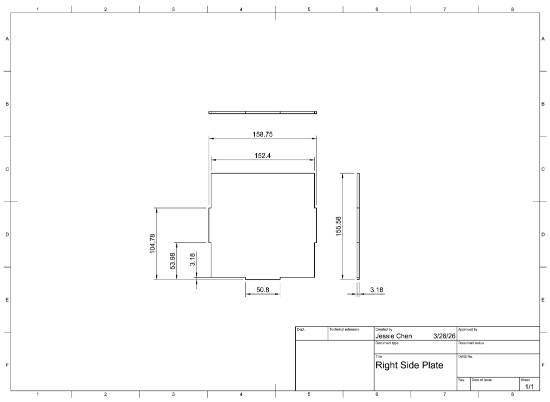


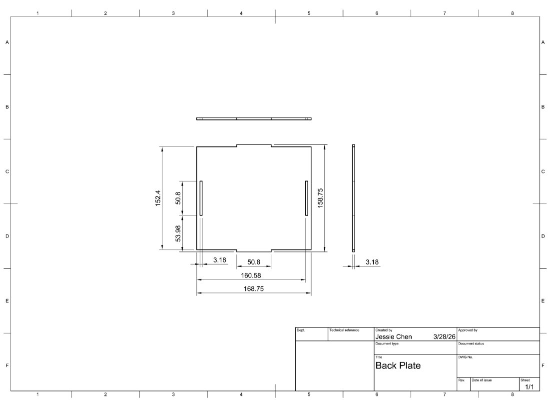
Box: laser cut top, base, left, right and back plates from sheets of wood



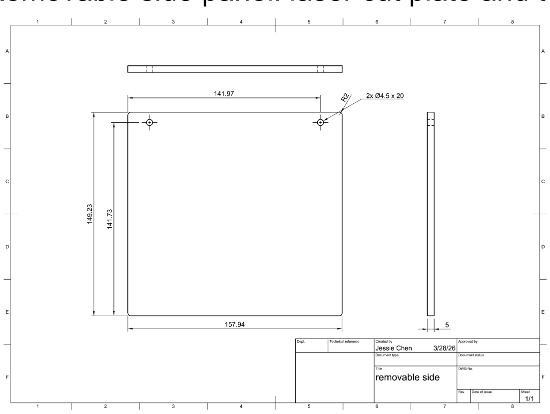


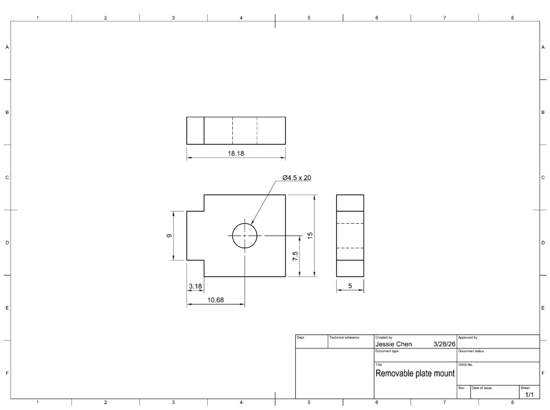




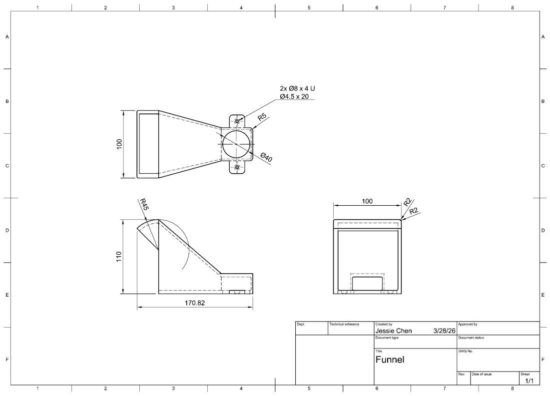


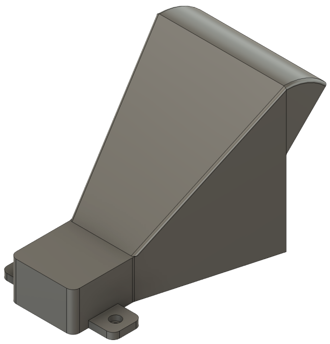
Removable side panel: laser cut plate and two mounts from sheets of wood



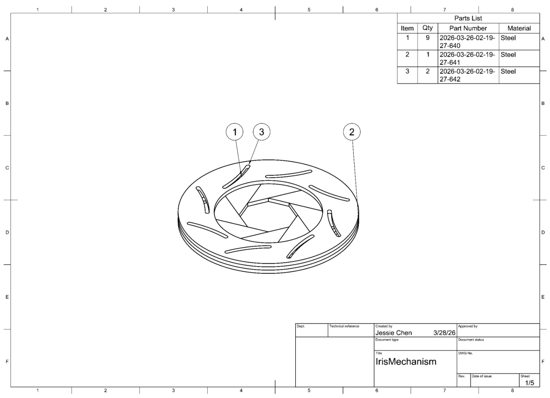


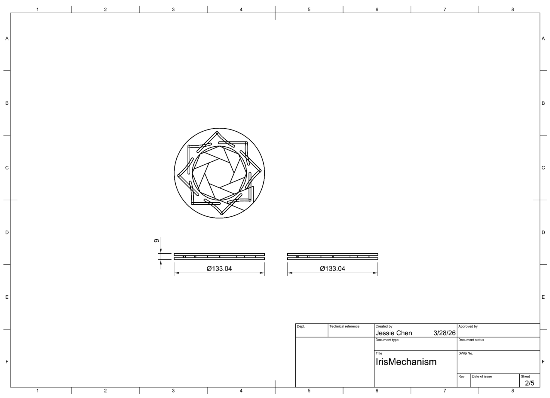
Funnel: 3D print

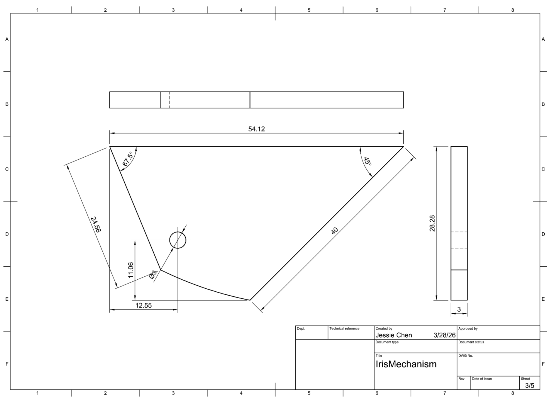


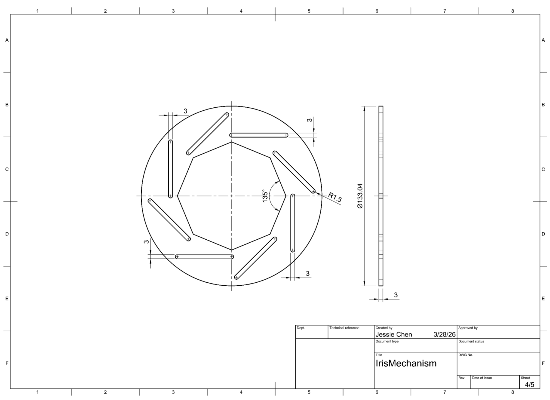


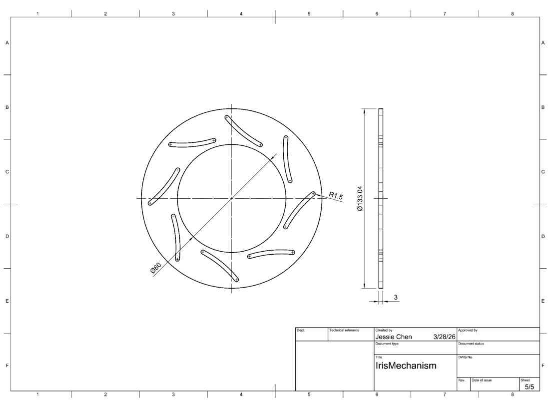
Iris: 3D print all parts

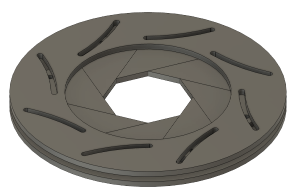




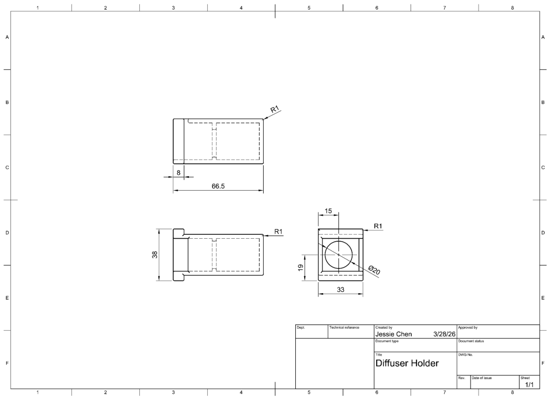


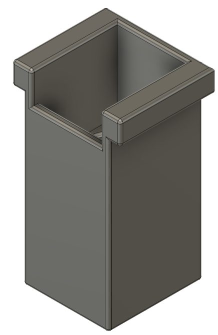


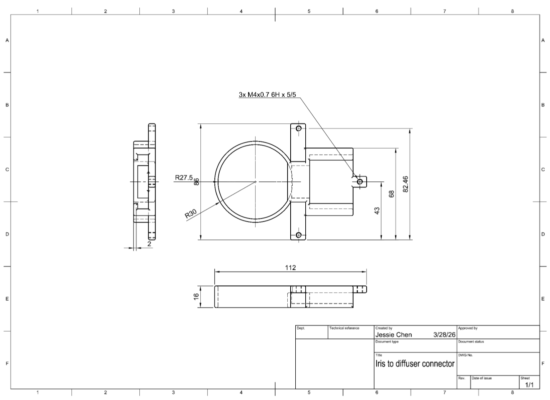


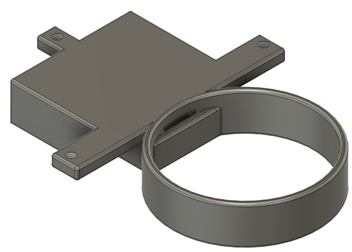


Chemical diffuser: separately 3D print diffuser holder and connector, purchase cotton balls from Amazon









Assembly

Overall trap:

1. Press fit the five plates of the box, orient left, right and base plates so that the smooth edges face the edge with two shorter slots and one longer slot of the top plate
2. Glue one M4 screw to the hole on each of the mounts, orient the mounts so that the screws aim outward the box and glue the extrusions of the mounts to the shorter slots on the top plate, hang the removable side panel on the screws
3. Screw the funnel from above the top plate with M4 screws and bolts, letting the opening of the funnel face the removable side panel
4. Screw the diffuser connector from below the top plate with M3 screws and bolts, aligning the circular ring of the connector to the bottom of the funnel

5. Place one cotton ball in the diffuser holder and slide the holder onto the connector

Iris: Place the blades on the bottom plate, orient the blades so that the iris is at maximum opening, then put on the top plate and adjust the opening to required size

Design Test

First test: Iris Mechanism

The iris consists of the adjustable opening on the top of the box. We are testing the functionality of the trap with this, as it allows the chemical to be diffused into the environment and opens to attract the SLFs into the box. For this part, the iris should be large enough to allow the SLFs through. The success criteria for this test were that the iris opens to a diameter greater than the maximum SLF size and operates smoothly without mechanical failure. The iris mechanism itself worked as the opening is greater than the size of an SLF. The iris opens to 80mm in diameter, and the size of an SLF is 12.7 - 25.4mm. We repeatedly opened/closed the iris to test the durability and consistency. However, some components of the iris broke when putting the parts together, and we were not able to attach the iris to the box ceiling. We have to see the iris integration in our next prototype. Chemical diffusion effectiveness was not tested in this iteration, but it is intended for future experiments.

Second test: Integration of diffuser attachment

The attachment holds the methyl salicylate chemical. This test is to see if the attachment can be seamlessly integrated into the assembly without interfering with how this part and the iris will operate. The success criteria were that the diffuser can be attached securely and operate without interfering with the iris mechanism. The diffuser holder was able to slide in and out of the diffuser attachment. However, the part’s connector to the iris currently interferes with the assembly of the iris inside the box due to spatial conflict. Because of this, we determined that this part of the connector should be removed to ease the iris integration. In the next prototype, the redesign will not interfere with the iris.

Third test: Removable side panel

This is the panel that opens/closes the trap box and allows the user to access the inside of the box when the sticky trap or the chemical needs to be replaced. The success criteria were that the panel fully encloses the trap without gaps and can be easily removed and reattached for maintenance. Currently, this panel consists of two M4 screws on the top corners so that the side panel can be removed and placed back on. The panel can be opened and closed successfully for access. However, there is a gap at the bottom of the box where the chemical can be diffused out, and SLFs could also escape the trap. We will add two additional M4 screws on the bottom corners to ensure a secure enclosure of the trap under operations.

Success Criteria

Our project aims to design a trap to lure and kill adult SLF. The trap uses methyl salicylate as a chemical attractant combined with sticky surfaces to capture adult SLF in outdoor environments.

- The trap should capture at least 100 adult SLF per box before requiring servicing. This can be measured by counting trapped insects after a two-week field deployment.
- The trap should remain structurally intact and functional after at least 14 consecutive days of outdoor exposure, including rain and wind. Success is measured by inspecting for deformation, material or seal failure, or adhesive degradation after the deployment period. Maintaining function is a high priority; exceeding 14 days is a low priority.
- A user should be able to replace both the chemical diffuser and sticky trap components in under 2 minutes, given that replacement supplies are on hand. This will be timed with a stopwatch across 3 trials with an unfamiliar user; the average must be under 2 minutes. Faster replacement times are a high priority.
- The methyl salicylate diffuser should provide a steady, detectable release for at least 14 days without refilling. This will be measured by weighing the diffuser at the start and end of the two-week period and confirming a consistent daily mass loss of at least 0.1g/day. Longer duration is a high priority.

For the end-of-semester exhibition, we will demonstrate the serviceability criterion with a live demonstration of removing and replacing the chemical diffuser and sticky trap components using the removable side panel mechanism in 2 minutes.